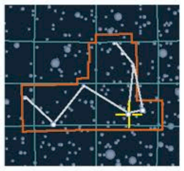
TRIPLE STAR

Beta Monocerotis



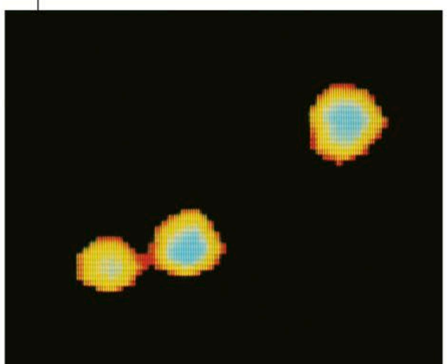
DISTANCE FROM SUN 700 light-years

MAGNITUDE 5.4

SPECTRAL TYPE B2

MONOCEROS

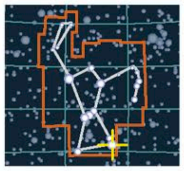
Beta (β) Monocerotis is a triple star system with components A, B, and C. The BC pair orbits each other with a period of about 4,000 years, and A orbits the BC pair with a period of about 14,000 years. The system is unusual because the three stars are so similar. All are hot, blue-white B-type stars, each more than 1,000 times as luminous and six times as massive as the Sun. All three stars also exhibit the same rotation speed and have circumstellar disks.



COMPUTER-ENHANCED OPTICAL IMAGE

TRIPLE STAR

Rigel



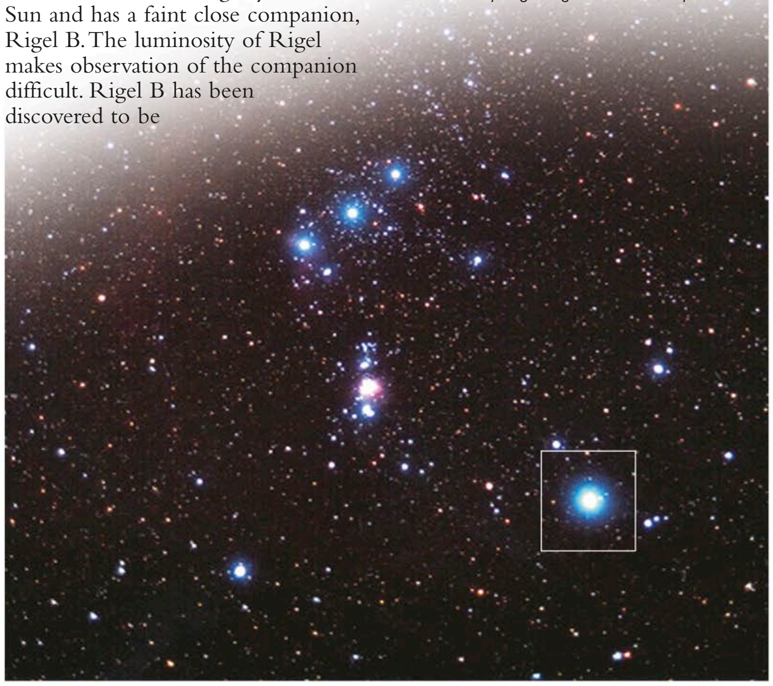
DISTANCE FROM SUN 860 light-years

MAGNITUDE 0.1

SPECTRAL TYPE B8

ORION

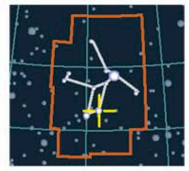
Rigel is a blue supergiant star shining 40,000 times more brightly than the Sun and has a faint close companion, Rigel B. The luminosity of Rigel makes observation of the companion difficult. Rigel B has been discovered to be a double star itself. It consists of two faint B-type main-sequence stars, Rigel B and Rigel C, separated by about 2.5 billion miles (4 billion km) and orbiting each other in an almost circular orbit. By contrast, the separation between the bright supergiant and the BC pair is over 190 billion miles (300 billion km).



BRILLIANT GIANT
Rigel is the brightest star in the constellation Orion and the 7th-brightest star in the night sky. Rigel B and C, its companion stars, are obscured by Rigel's great luminosity.

DOUBLE STAR

Beta Lyrae



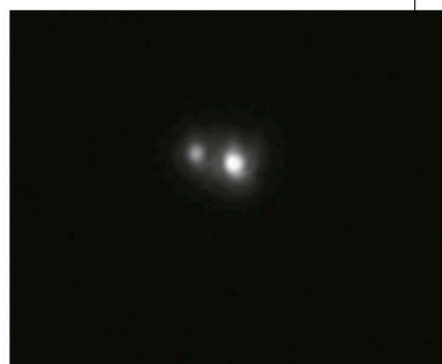
DISTANCE FROM SUN 880 light-years

MAGNITUDE 3.5

SPECTRAL TYPE B7

LYRA

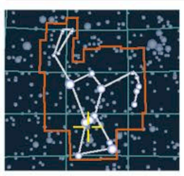
Beta (β) Lyrae, or Sheliak, is the prototype of a class of eclipsing binary stars known as Beta Lyrae stars or EB variables (see p.274). The brightness of the system varies by about one magnitude every 12 days 22 hours and is easily visible to the naked eye. Beta Lyrae's component stars are contact binaries and are so close together that they are greatly distorted by their mutual attraction. Material pouring out of the stars is forming a thick accretion disk.



CLOSE BINARY

QUINTUPLE STAR

Sigma Orionis



DISTANCE FROM SUN 1,150 light-years

MAGNITUDE 3.8

SPECTRAL TYPE O9

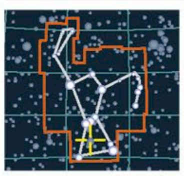
ORION

Sigma (σ) Orionis is a quintuple system containing four bright, easily visible stars and one fainter component, with the brightest being a close double. The two main stars, A and B, are more than 30,000 times as luminous as the Sun and have a combined mass more than 30 times greater than the Sun. The AB pair is one of the more massive binary systems in the Milky Way. It is in a stable orbit, but the C, D, and E stars are not, and gravitational forces may well throw them out of the system in the future.

SIGMA ORIONIS'S FOUR BRIGHT STARS

QUADRUPLE STAR

Theta Orionis



DISTANCE FROM SUN 1,800 light-years

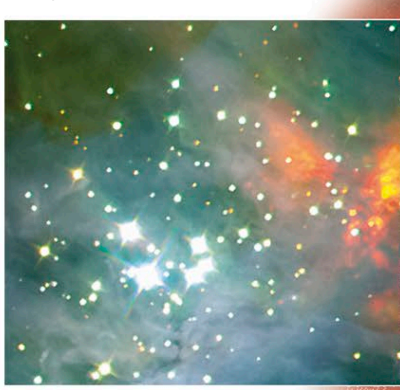
MAGNITUDE 4.7

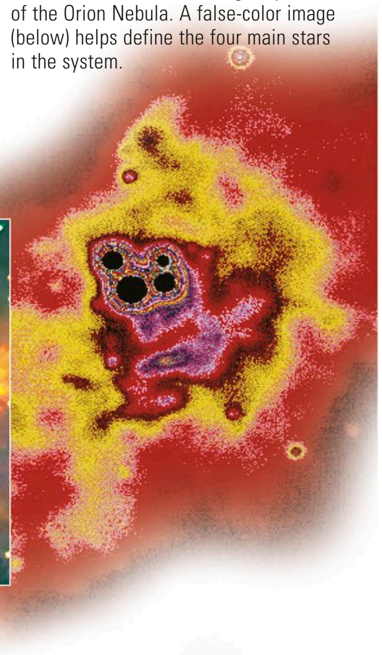
SPECTRAL TYPE B

ORION

Theta (θ) Orionis, perhaps better known as the Trapezium, appears to the naked eye to be a single star but is revealed by any telescope to be a quadruple system. Theta Orionis provides much of the ultraviolet radiation that illuminates the Orion Nebula (see p.241). All four stars are hot O- and B-type stars, the largest being Theta-1 C, with 40 times the mass of the Sun; about 200,000 times its luminosity; and a temperature of 72,000°F (40,000°C). Theta-1 C is the hottest star visible to the naked eye. Theta-1 A is an eclipsing double star with an additional companion, Theta-1 D is a double star, and Theta-1 B is an eclipsing binary star with a companion double (making it quadruple in itself). The multiple star forms the core of an open cluster containing hundreds of young stars.

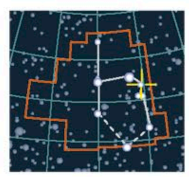
THE TRAPEZIUM GROUP
The stars of Theta Orionis light up the center of the Orion Nebula. A false-color image (below) helps define the four main stars in the system.





DOUBLE STAR

Epsilon Aurigae



DISTANCE FROM SUN 2,040 light-years

MAGNITUDE 3

SPECTRAL TYPE A8

AURIGA

The hot giant Epsilon (ε) Aurigae, or Almaaz, is an eclipsing binary star. Unusually, its eclipse lasts for 2 years, suggesting that the system is huge. The giant star is being eclipsed by something far bigger than itself, but exactly what is uncertain. One theory is that Epsilon Aurigae's companion is an unseen binary system surrounded by a huge, dusty ring, through which the bright star shines through this ring during an eclipse.

DISTANT BINARY
Dwarfed in this image by its celestial neighbor Capella, Epsilon Aurigae experiences its strange eclipses every 27 years.